MANAKULA VINAYAGAR INSTITUTE OF TECHNOLOGY

An Autonomous Institution

Affiliated to Pondicherry University, Approved by AICTE, New Delhi.

Accredited by NAAC with 'A' Grade

Kalitheerthalkuppam, Puducherry – 605 107.

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

INTERNAL EXAMINATION - I

Subject / Code : DISCRETE MATHEMATICS/CSPC401

Year/Semester : II / IV

Dept : CSE-A

Date: 09/02/2026

Time: 8.50 am to 10.30 am

Max.Marks: 50

[Bloom's Taxonomy: K1- Remember, K2- Understand, K3-Apply, K4-Analysis]

[illegible]	[illegible]	[illegible]
CO1	Explain sets and relations, demonstrate function properties, and apply Cantor's arguments to prove cardinality distinctions.	Apply(K3)
CO2	Apply proof techniques, mathematical induction, explain number-theoretic algorithms, and solve modular arithmetic problems.	Apply(K3)

PART – A

Answer ALL the questions **(10*2=20)**

1. Define Cartesian product of sets. (CO1:K1)
2. Define Relation. (CO1:K1)
3. Check whether the following relation $R = \{(0,0), (1,1), (1,3)(2,2), (2,3), (3,1), (3,2), (3,3)\}$ is an Equivalence relation on a set $\{0,1,2,3\}$. (CO1:K2)
4. Explain injective and surjective functions. (CO1:K2)
5. Determine whether the following relation $R = \{(0,0), (0,2), (2,0)(2,2), (2,3), (3,2), (3,3)\}$ is a function. (CO1:K2)
6. Explain POSET. (CO1:K2)
7. Define Countable. (CO1:K1)
8. Write any two methods of Mathematical Proofs. (CO2:K1)
9. Define division algorithm. (CO2:K1)
10. Find the GCD of 1575 and 231 using Euclidean Algorithm. (CO2:K3)

PART-B

Answer ALL the questions **(3*10 = 30)**

11. (i) Let R denote the set of real numbers, and let the function $f: R \rightarrow R$ be defined by $f(x)=x^3-2$. Prove that f is injective and surjective. Also, determine the inverse function f^{-1}. (5)(CO1:K3)
    (ii) Prove that the relation greater than or equal to ($\geq$) is a partial ordered set on the set of integers. (5)(CO1;K3)
12. (i) State and Prove the Power set theorem. (4)(CO1;K3)
    (ii) State and Prove Cantor's diagonal argument. (6)(CO1:K3)
13. (i) Prove that $\sqrt{2}$ is an irrational number. (4)(CO2: K3)
    (ii) Prove that $1^3 + 2^3 + \ldots + n^3 = \left(\frac{n(n+1)}{2}\right)^2$ for all positive integers n by using mathematical induction method. (6)(CO2:K3)

********** ALL THE BEST **********